

Statistics: To Know Concepts Checklist, IES FSV UK, 2025

This “checklist” is organized in the same order as the lectures. It is meant as a *concept map* $\approx$ what you should know and be able to use, it is not a formula sheet.

Study tip: For each bullet above, you should be able to (i) state the definition or theorem or concept, (ii) apply it to compute a probability/estimator/CI/test, and (iii) interpret the result in words.

1 Introduction

- What statistics is for (especially in economics): decision-making under uncertainty; separating signal from noise; linking models to data.
- Basic workflow: intuition $\rightarrow$ formal definition $\rightarrow$ computation $\rightarrow$ interpretation.
- Course meta-structure (how later parts fit together): random variables $\rightarrow$ expectations/variation $\rightarrow$ random samples $\rightarrow$ estimation $\rightarrow$ confidence intervals $\rightarrow$ testing.
- Descriptive vs. inferential statistics; parameter vs. statistic; data-generating process idea.
- Data summaries and visualization; Sample space and events; Counting techniques (permutations, combinations); Conditional probability; Bayes’ theorem; Independence of events.

2 Random Variables

- Random variable (RV): definition as a numerical function on the sample space; events written using an RV (e.g. $\{X \leq a\}$).
- Types of RVs: **discrete** vs. **continuous** (and what changes in how we compute probabilities).
- Distribution of an RV:
 - pmf $p(x)$ for discrete RVs; pdf $f(x)$ for continuous RVs; relation to probabilities.
 - CDF $F(x) = P(X \leq x)$, its properties, and how to use it to compute probabilities.
- Core named distributions and how to work with them (support/shape/parameters, computing probabilities, using tables/software):
 - Discrete: **uniform (finite)**, **binomial**, **Poisson** (incl. basic examples).
 - Continuous: **uniform**, **exponential**, **normal**; **standardization** for normal.
- Quantiles: definition via the CDF; reading/interpreting z_α -type notation.
- Transformations of RVs:
 - Discrete transformations: push-forward of probabilities by mapping values.
 - Continuous transformations: change-of-variables idea in one dimension; computing the resulting pdf/cdf.
 - Probability integral transform (PIT): the main idea behind simulation.

3 Multivariate Distributions and Transformations

- Random vectors / pairs of RVs: motivation (e.g. income vs. consumption); joint behavior vs. marginal behavior.
- Joint distributions:
 - Joint pmf/joint pdf; joint CDF $F_{X,Y}(x, y)$ and how it encodes probabilities of rectangles.

- Fubini theorem as the justification for changing order of summation/integration and for marginalization.
- Marginal distributions: obtaining $f_X(x)$ and $f_Y(y)$ by summing/integrating out the other variable.
- Independence:
 - Definition for discrete/continuous cases (factorization of the joint).
 - How independence simplifies probability calculations.
- Multivariate transformations:
 - Change-of-variables with a Jacobian determinant (conceptually and computationally).

4 Expectations

- Expectation $E[X]$:
 - Discrete vs. continuous definitions; interpreting expectation as a long-run average / center of mass.
 - When $E[X] = \pm\infty$ vs. when $E[X]$ does not exist.
 - Expectation of functions: expectation of $g(X)$ without first finding the distribution of $g(X)$.
- Core properties: linearity, monotonicity, and scaling/translation rules.
- Moments:
 - Ordinary and central moments; variance as the second central moment.
 - Existence of moments; Liapunov inequality (what it says and why it matters).
- Moment generating function (mgf):
 - Definition, when it exists, and how to compute moments from derivatives.
 - Key properties (e.g. mgf of sums of independent RVs).
- Variance and standard deviation: computation rules, interpretation as variability/risk.
- Skewness and kurtosis: what they measure and qualitative interpretation.
- Inequalities and concentration:
 - Chebyshev inequality and how to use it for probability bounds.
 - “Beyond Chebyshev” intuition; the 3σ rule for the normal distribution.
- Covariance and correlation:
 - Covariance definition; variance/covariance of linear combinations.
 - Independent vs. uncorrelated.
 - Correlation coefficient $\rho_{X,Y}$; notion of Schwarz (Cauchy–Schwarz) inequality and implications.

5 Conditional Distributions

- Conditional distributions:
 - Discrete: $P(X = x \mid Y = y)$ and conditional pmf.
 - Continuous: conditional density $f_{X|Y}(x \mid y)$; relation to joint and marginal densities.
- Conditional CDFs and how independence appears through conditioning.
- Applications/structures:
 - Thinning-type ideas (conditioning and random selection leading to new distributions).

- Mixtures of distributions (a distribution built by randomizing parameters or regimes).
- Conditional expectation $E[X | Y]$:
 - Interpretation as the best mean-squared predictor of X given Y (conceptual).
 - Law of iterated expectation (total expectation): $E[E[X | Y]] = E[X]$.
 - Law of total variance (iterated variance): decomposition into “within” and “between” components.
 - Substitution theorem (using measurable transformations inside conditional expectations).

6 Selected Families of Distributions

- Bernoulli trials / alternative (0–1) distribution and its role as a building block.
- Discrete families (setup, parameters, typical applications, key properties):
 - Binomial (incl. examples), geometric (incl. examples), negative binomial (incl. examples).
 - Hypergeometric (sampling without replacement; properties and examples).
 - Poisson (counting model; properties and applications).
- Continuous families:
 - Uniform, exponential (incl. memoryless property).
 - gamma and χ^2 distribution, their relation.
 - Normal distribution (review + key properties); lognormal distribution and why it matters in economics/finance.
- Random vectors:
 - Expectation of random vectors; variance–covariance matrix; correlation matrix. Ideas and basic notion.
 - Bivariate normal distribution (Ideas and basic notion).

7 Random Samples

- Random sample and statistic: definition; sampling distribution as the distribution of a statistic.
- Sample mean and sample variance:
 - Basic properties (expectation/variance; why the denominator differs for variance estimators).
 - For normal samples: distributional results and key independence facts.
- t distribution:
 - How it arises from a normal sample with unknown variance; main properties and usage.
- F distribution: definition (ratio of scaled χ^2 variables) and typical use-cases.
- Order statistics:
 - Definition (min/max, k -th order statistic) and why their CDFs are useful.
- Modes of convergence:
 - Convergence in probability, almost surely, in distribution; key differences and intuition.
 - Implication chain (a.s. $\Rightarrow$ in prob. $\Rightarrow$ in dist.) and counterexamples/limitations.
- Laws of Large Numbers:
 - Weak vs. strong LLN (statements, conditions, interpretation).

- Central Limit Theorem (CLT):
 - Statement and standardization; practical meaning for approximations.
 - Examples; continuity correction for approximating discrete distributions by normal.
 - Awareness of “more CLTs” (extensions/conditions) as referenced in the slides.

8 Point Estimation

- Statistical inference setup: data, model family $\{f(x, \theta) : \theta \in \Theta\}$, unknown parameter(s).
- Estimator vs. estimate; statistic as a function of the sample.
- Quality criteria for estimators:
 - Bias and unbiasedness; variance.
 - Mean squared error (MSE) and bias–variance trade-off.
 - Consistency: definition and how to check it (incl. “test of consistency” using Chebyshev).
- Fisher information:
 - Score function $J(X, \theta) = \partial_\theta \log f(X, \theta)$ and $I(\theta) = E[J^2]$.
 - Information in a sample: $I_n(\theta) = nI(\theta)$.
- Rao–Cramér lower bound (CRLB): lower bound on variance of unbiased estimators; interpretation.
- Efficiency and (relative) efficiency: definition and how to compare unbiased estimators.
- Construction methods:
 - Method of Moments (MoM): matching sample and population moments; solving (possibly multiple) equations.
 - Maximum Likelihood Estimation (MLE): likelihood, log-likelihood, first-order conditions; interpreting MLEs.
 - Least Squares (LS): estimator defined by minimizing squared residuals (connection to basic regression ideas).

9 Interval Estimation

- Confidence interval (CI):
 - Definition via coverage probability $P_\theta(L(X) \leq \theta \leq U(X)) = 1 - \alpha$.
 - Lower/upper confidence bounds; what “random interval” means.
 - Distinction between probability CI $(L(X), U(X))$ and realized sample CI $(L(x), U(x))$.
- Quantile notation used in CIs: z_α , $t_{\alpha, \nu}$, and analogues for χ^2 and F .
- Classic normal-model CIs:
 - Mean of $N(\theta, \sigma^2)$ with known σ^2 (z-based CI).
 - Mean with unknown σ^2 (t-based CI using S^2 and independence from $\bar{X}$).
 - Variance parameter (chi-square based CI; effect of known vs. unknown mean).
- Large-sample (approximate) CIs based on asymptotic normality and Fisher information:
 - Efficient estimators and asymptotic standard normal scaling.
 - Approximate CI built around the MLE using $I(\hat{\theta})$.
 - Example: Bernoulli(p) CI using $\hat{p} = \bar{X}$ and plug-in standard error.

10 Testing Statistical Hypotheses

- Hypothesis testing framework:
 - Null vs. alternative; simple vs. composite hypotheses.
 - Test function / rejection rule; critical region.
- Errors and performance:
 - Type I and Type II errors; significance level α .
 - Size vs. level of a test; power function and its interpretation.
- Generalizations:
 - Constructing critical regions from a chosen test statistic and its null distribution.
- p -values: definition, computation, and interpretation (what they are and are not).
- Link between tests and confidence intervals: inversion principle (rejecting values outside a CI).
- Standard test catalog to know how to set up and execute:
 - One-sample mean tests: σ^2 known (z-test), σ^2 unknown (t-test).
 - One-sample variance tests (chi-square based).
 - Two-sample mean tests (independent samples; equal/unequal variance variants).
 - Paired tests (matched pairs) vs. two-sample tests (why the setup changes).
 - Large-sample tests based on CLTs (approximate tests when n is large).